Calibration Scope in Mechanism-Level SHAP Audits: A Controlled Study

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***Abstract*—Predictive maintenance requires reliable fault predictions and understandable model behavior. Mechanism-level audits use Shapley additive explanations (SHAP) to compare feature contributions with expected fault signatures. The practical question is how their warning thresholds transfer across refitted models and whether attribution warnings identify predictive harm when operating conditions change. Here we show that warning interpretation depends on the prediction output and calibration population, using split-conformal ranks and 1,200 controlled model pairs with exact interventional SHAP. At nominal 5% calibration on one model, same-family refits with identical decisions but different probabilities produced warning rates up to 9.6%; hard-output score laws agreed under the same background and batch law. At matched empirical 5% eligible-model warning rates, pre-shift localization varied by learner and score design, while a performance screen using the known context input and labels matched or exceeded attribution-warning detection. Process replays exposed saturated decisions, while measured vibration under load transfer revealed healthy-row failures sensitive to the binary decision rule. These exploratory findings support specifying the audited output and calibration population, then assessing predictive harm separately from attribution concentration.**

***Index Terms*—SHAP, explanation evaluation, conformal calibration, distribution shift, model multiplicity**

I. Introduction

A. Background

Predictive maintenance uses sensor measurements to diagnose equipment faults and support maintenance decisions. Engineers need to assess both predictive accuracy and whether a model uses features consistent with the fault under investigation [1], [2]. Models can exploit associations between faults and acquisition conditions that change across operating environments [3], [4].

Shapley additive explanations (SHAP) assign feature contributions to a selected model output [5]. A mechanism-level audit aggregates these contributions for each fault type and compares them with a supplied map of expected features. Attribution outside this map may reflect unintended reliance, incomplete engineering knowledge, or another genuine fault mechanism. Such attribution does not by itself establish physical causation or predictive harm [6]. Interpreting an audit warning therefore requires a defined reference population and a clear predictive target.

B. Research Gap

Explanation variability across similarly performing models is established [7], and conformal calibration controls warning rates under its stated exchangeability assumptions [8]. This leaves two questions for mechanism-level audits: how much warning rates change across decision-equivalent refits, and whether attribution warnings identify predictive loss under changed operating conditions. Identical class decisions can conceal different probabilities, and a warning about unexpected attribution need not identify a model that loses accuracy. Practitioners must therefore distinguish explanation variation, input reliance, and predictive harm.

C. Contributions

Here we examine these distinctions using 1,200 controlled model pairs and exact interventional SHAP, with process simulations and measured vibration data as diagnostic illustrations. The study makes three contributions:

- Same-family refits with identical decisions produce elevated probability-audit warning rates; an exact hard-output control identifies the background and batch-law conditions under which transfer is automatic.
- Matched-rate comparisons distinguish unknown-feature warnings, known-context localization, and performance screening; score-design ablations quantify sensitivity to background and normalization.
- Process replays explain saturated binary decisions, while measured load transfer exposes healthy-row failures and sensitivity to the binary decision rule.

These findings support specifying the audited output and reference population, then assessing predictive harm separately from attribution concentration. The measured-load pilot illustrates prediction failure; it does not establish calibrated warning sensitivity across independent physical assets.

II. Related Work

A. Explanation Evaluation

SHAP assigns contributions to a selected scalar output [5]. Its interventional and observational value functions answer different questions [9]–[11]; neither establishes physical causation [6]. Expected-feature share is related to relevance mass accuracy [12].

Permutation importance [13] evaluates predictive dependence. SHAP-XRT [14] tests local conditional independence, whereas our score tests batch-level excess against a mechanism map. Our permutation-performance comparator is not SHAP-XRT or a conditional-independence test. Explanation Shift [15] compares explanation distributions; here, warning rates and localization are evaluated against measured predictive consequences.

B. Model Multiplicity

Underspecification allows similarly performing fits to behave differently in deployment [16]. Model class reliance bounds importance across well-performing models [17]; predictive multiplicity concerns conflicting decisions [18]. Brunet et al. [7] establish explanatory multiplicity under model indeterminacy, while Krishna et al. [19] study disagreement between explanation methods. Variable importance clouds [20] and Rashomon-set attribution consensus [21] describe variation across good models; Rashomon capacity explicitly measures probability-level multiplicity [22]. Our contribution is a controlled evaluation of how this variation affects a particular mechanism-level warning threshold, with identical hard decisions over the complete finite domain. It is not a general discovery of explanation instability.

C. Harmful Shift

Rabanser et al. [23] study dataset-shift detection and malignancy; Podkopaev and Ramdas [24] provide sequential risk-monitoring guarantees. Our realized F1-loss labels are retrospective diagnostics. Context-use warnings do not inherit a harm-detection guarantee.

D. Conformal Calibration

Split-conformal ranks provide marginal control under exchangeability [8]. Mondrian methods condition calibration on categories [25], [26]; conditional conformal methods extend the range of conditional targets [27], and weighted methods address specified covariate shifts [28]. These results motivate stratification, not a new calibration theorem. We compare fixed overlap bins with bins learned from source covariates, and distinguish marginal control from calibration-conditional uncertainty [26], [29].

E. Maintenance Applications

Maintenance-XAI work motivates inspecting expected fault signatures [1], [2], [30]. Time-series XAI [31] and TimeSHAP [32] address temporal explanations. Our process study instead treats selected timestamps as tabular inputs and evaluates neither temporal localization nor an industrial deployment intervention.

III. Audit Framework

A. SHAP Attribution

Table I fixes the terminology. For a scalar model output $h(x)$ and feature subset J, our interventional value function is

$$v_h(J;x) = \mathbb{E}_{X' \sim \widehat{P}}[h(x_J, X'_{-J})]. \tag{1}$$

TABLE I
Terms used throughout the audit. A reference may be a fixed function or a specified distribution of independently trained functions.

Symbol	Meaning
f, g, h	Reference predictor, candidate predictor, and a generic scalar predictor
E_m	Expected features supplied for mechanism m
$\mathcal{I}_m$, $\mathcal{F}$	Rows with mechanism m; all recorded failure rows
C, ϵ	Context input and its bit-flip probability relative to Z_0
H_i	Realized mean F1 loss from intact to independent context
Q_b, c_α	Base-bin reference and residual-calibration cut
A_h, L_h	Maximum excess; excess localizing the specified context
Eligible fit	Mechanism-only model meeting the stated full-state gate
Harm-positive	Candidate with mean F1 loss at least 0.05 under independent context

The selected inputs come from x; missing inputs come jointly from one background row. For d inputs, the attribution is

$$\phi_j^h(x) = \sum_{J \subseteq [d] \setminus \{j\}} w_d(J)\, \Delta_j^h(J;x), \tag{2}$$

$$\Delta_j^h(J;x) = v_h(J \cup \{j\};x) - v_h(J;x), \tag{3}$$

$$w_d(J) = \frac{|J|!(d-|J|-1)!}{d!}. \tag{4}$$

The coefficient averages the prediction change from adding feature j over feature orders.

B. Audit Scores

For a nonempty row group A, define mean absolute attribution and normalized share as

$$a_j^h(A) = \frac{1}{|A|} \sum_{x \in A} |\phi_j^h(x)|, \qquad s_j^h(A) = \frac{a_j^h(A)}{\sum_k a_k^h(A)}. \tag{5}$$

The normalization is a ratio of group means, not a mean of per-row ratios. For mechanism m, define the outside-set excess

$$S_m = \max_{j \notin E_m} \{ s_j(\mathcal{I}_m) - s_j(\mathcal{F} \setminus \mathcal{I}_m) \}. \tag{6}$$

The model superscript is omitted when clear. Healthy rows belong to neither comparison group, so the score does not directly count their errors. Empty groups or zero attribution make it unavailable.

Let $A_h(X) = \max_m S_m^h(X)$ for batch X. For localization, retain S_m^h only when its maximizing outside feature is C, then maximize over those mechanisms; call the result $L_h(X)$, or $-\infty$ if none qualifies. Ties use a fixed feature order. Localization is an attribution event pointing to C, not a finding of transfer harm.

C. Mechanism Overlap

An outside feature can be a genuine co-occurring mechanism rather than a shortcut. Let $Z_0 = Z_1 \sim \text{Bernoulli}(\pi)$ and independent $Z_2 \sim \text{Bernoulli}(\rho)$, with perfect predictor $h = Z_0 \vee Z_1 \vee Z_2$ and $E_m = \{Z_m\}$. Exact interventional SHAP gives

$$S_0 = \left[\frac{3}{6+4\rho} - \frac{\pi}{6-2\pi} \right]_{\pi=\rho=0.05} = 0.4754, \tag{7}$$

TABLE II
POPULATION-REFERENCE AUDIT RECIPE. FIXED-FUNCTION CALIBRATION REPLACES INDEPENDENT FITS BY BATCHES SCORED WITH ONE FROZEN FUNCTION.

Inputs: output, expected-feature sets, background, cohorts, eligibility gate, training/batch law, partition, and α.
1. Freeze design. Specify the output and whether the warning concerns attribution, reliance, or predictive loss. Healthy-row errors require an error statistic; cohort coverage alone is insufficient.
2. Build bank. Declare training data, learner, randomness, and eligibility. Draw independent fits and one independent audit batch per accepted fit; apply the frozen score.
3. Calibrate. In the prespecified bin, use rank $k = \lceil (M_b + 1)(1-\alpha) \rceil$. Nineteen units make a 5% cut finite, without providing precise realized size.
4. Audit. Score the candidate under the same design. Abstain on missing cohorts, zero denominators, or $k > M_b$.
Output: warn on strict cut exceedance; localize an eligible maximizing feature. Report reference scope, stratum counts, abstention, and bank uncertainty. Validate proposed features with labels and a justified intervention, including healthy/faulty errors under independent acquisitions.

maximized by genuine mechanism Z_1. This credit sharing under dependence [6], [11] affects the warning score without implying harmful reliance. The full enumeration accompanies the technical materials.

IV. METHODS

A. Audit Procedure

Fixed-model and population calibration separated batch variation from fitting variation. Localization and predictive consequences were assessed separately because attribution concentration need not imply harm. Output, feature map, cohorts, background, and reference law were specified before scoring (Table II). Unsupported map relationships remained unresolved; incomplete-map bounds are retained in the technical materials.

B. Output Choice

The hard-output control separated probability variation from classification disagreement. Let $d_h(x) = \mathbf{1}\{h(x) \geq 0.5\}$. If $d_f = d_g$ on the entire input domain, coalition values, SHAP attributions, and derived scores agree for those decisions under the same background; transfer laws agree under the same batch law (Eqs. (1) and (2)). Matching decisions implies neither equal probabilities nor equal log-odds explanations.

C. Fixed-Model Calibration

Fixed-model calibration held fitting constant to isolate batch variation. Observed overlap is $O(X) = \widehat{\Pr}(Z_1 = 1 \mid Z_0 = 1)$. The base-bank $(1-\alpha)$ rank cut Q_b was frozen per overlap bin before target calibration. Define $T_h(X) = A_h(X) - Q_{b(O(X))}$. From M independent target batches scored by fixed f, let c_α be the kth ordered residual, $k = \lceil (M+1)(1-\alpha) \rceil$, or $+\infty$ if $k > M$. A candidate warns when $T_g > c_\alpha$ and localizes context when $L_g > Q_b + c_\alpha$.

Exchangeability of calibration and future reference scores gives the standard rank bound [25], [26]

$$\Pr\{T_f(X_{\text{new}}) > c_\alpha\} \leq \frac{M+1-k}{M+1} \leq \alpha. \tag{8}$$

This covers self-warnings for f, not refits. A finite global offset Q cancels after residual calibration. Missing cohorts or zero attribution denominators make scores unavailable; base/global banks encode these as $+\infty$ and conditional residuals as $-\infty$. Invalid tests and unsupported bins abstain.

D. Population Calibration

Population calibration sampled eligible fitted functions to incorporate fitting variation. Each independent fit supplied one independent target batch and one score. Global or within-bin ranks were applied without residual re-calibration; repeated batches from one fit cannot represent fitting variation.

Joint exchangeability of eligible fit/batch pairs gives marginal rank control for that training procedure; within-bin ranks have the Mondrian target [25], [26]. This averages over fitting and batch variation, without extending control to another learner or physical map.

For i.i.d. scores, individual 95% order-statistic tolerance bounds [26] give warning-rate upper bounds of 6.89% ($M = 400$) and 7.72% ($M = 200$) at nominal 5%. The latter applies to global population calibration; within-bin cuts use their bin counts. These are not simultaneous or fixed-bank 5% guarantees.

E. Response Baselines

The flip-excess comparator retained the SHAP score's cohorts and feature search while replacing attributions by background-free prediction responses. Background and normalization differences were examined separately in the score ablations. For Boolean feature j, define

$$h_j^g(x) = |g(x) - g(x^{\neg j})|, \tag{9}$$

where $x^{\neg j}$ flips only j. The group mean of h_j^g replaced attribution share in the excess formula; global or observed-bin calibration matched the SHAP design. Independent uniform tie breakers ordered equal responses, including frequent zeros. SHAP retained strict ranks; warning-rate matching addressed realized size differences.

A targeted baseline uses $\rho_C = |X|^{-1} \sum_{x \in X} h_C^g(x)$, zero for every mechanism-only reference. Positive response rejects context invariance; zero-response ties use independent uniform ranks at 5%. This test knows C and establishes input use, not harm.

F. Harm Screening

To distinguish detection from feature identification, attribution warnings A_g and context localization L_g were evaluated against the same harm labels. The former searches outside features without knowing the context identity; the latter additionally requires C to maximize a mechanism contrast.

When C and outcome labels were available, a permutation-performance comparator assessed predictive consequences directly. Following permutation importance [13], [33], its score was

$$\Delta_{\text{perm}} = \text{F1}(g; X, Y) - \text{F1}(g; X^{\text{perm(C)}}, Y). \tag{10}$$

One uniformly random context-column permutation was generated per batch, with labels and other inputs fixed. Exact contingency counts were sampled by a multivariate hypergeometric draw. Eligible references implement the true decision function independently of C, so their performance drops are zero. Randomized population ranks ordered these ties; the same empirical warning-rate matching was applied to every screen. Unlike A_g, this comparator uses known C and labels.

Permutation preserves the marginal context prevalence. The designed $\epsilon = 0.5$ shift instead sets that prevalence to 0.5; the two interventions are not identical. A positive permutation loss measures sensitivity to an additional intervention on the current batch, not predictive damage already incurred before that batch.

G. Warning-Rate Matching

To compare localization at equal eligible-model warning rates, each score is first converted to a population-calibrated upper-tail rank:

$$p_h(X) = \frac{1 + \sum_{i \in \mathcal{B}_b} \mathbf{1}\{A_{f_i}(X_i) \geq A_h(X)\}}{|\mathcal{B}_b| + 1}, \tag{11}$$

where $\mathcal{B}_b$ contains the fit/batch pairs in the same overlap bin as X. The global rule uses every pair. Localization replaces A_h by L_h; flip scores use lexicographic score/uniform ranks. Unsupported strata and unavailable scores abstain. Applying $p_h \leq 0.05$ gives a prospective nominal rule under the specified population null.

For retrospective comparisons, a p-value boundary was selected using evaluated eligible controls so that the expected warning fraction equaled r. If $N_<$ and $N_=$ of their N p-values lie below and at that boundary, its warning probability is $w = (rN - N_<)/N_=$. The same mixture was applied to candidate localization or warning scores. Randomization contributes fractional expected detection at tied boundaries; it matches empirical size and does not define a prospective deployment threshold.

V. Experimental Design

A. Controlled Models

The Boolean design supplied known feature relevance and allowed exact enumeration, separating attribution differences from approximation error. Inputs were ordered (Z_0, Z_1, Z_2, C), with outcome $Y = Z_0 \vee Z_1 \vee Z_2$ and $E_m = \{Z_m\}$. Each stratum used independent 512-row training draws with uniform independent mechanism bits and 5% label flips. Paired references/candidates shared training data; seed schedules aligned samples across learner families. Context satisfied $C = Z_0 \text{XOR} \text{Bernoulli}(0.01)$. Tree depth equaled input count with minimum leaf five. Forests [33] used 64 such trees, square-root feature sampling, and bootstrap samples. Logistic regression used $C_{\text{reg}} = 1$, L-BFGS, and 1000 iterations.

Each learner/size stratum used 200 evaluated pairs. References used three mechanism inputs, matched OR decisions on all sixteen states, and ignored context; candidates also received context. Disjoint seeds supplied same-/other-family controls and 200 mechanism-only calibration fits, each scored on one separate target batch. Proxy controls replaced Z_0 by C to test a deliberately ineligible predictor.

Probability is the scalar output in the empirical attribution comparisons; the hard-output control checks equality on every state. Predictions use probability ≥ 0.5; feature-maximizer ties follow input order. Exact interventional SHAP enumerates all states/subsets with the empirical audit batch as background.

B. Batch Sampling

An audit batch contains $n = 1000$ or $10{,}000$ rows. Each batch draws regime probability $q \sim U(0.1, 0.4)$ and independent low/high mechanism probabilities from $U(0.01, 0.04)$/$U(0.1, 0.3)$. Each row draws one Bernoulli(q) regime; mechanism bits are conditionally independent before nesting. Nesting additionally activates Z_1 on Z_0 rows with probability u. Source batches use $u \sim U(0, 1)$ and target batches $u \sim U(0.8, 1)$. Each fixed reference has 1000 source-base, 1000 target-base, 400 target-calibration, and 200 untouched test batches. Source quantiles use no target outcomes.

Partitions comprise fixed bins $[0, 0.8), [0.8, 0.9), [0.9, 1]$ and source-only tertiles/quintiles, estimated from 1000 source batches per fixed function or one source batch per calibration fit. Fixed edges exploit the designed target law and serve as a target-informed reference, not a proposed data-driven partition. Global and source-derived partitions do not use those target-law edges.

C. Shortcut Localization

Paired context shifts isolate context association from model fitting and failure prevalence. Each candidate is frozen while its test context-flip probability changes to $\epsilon \in \{0.01, 0.10, 0.25, 0.50, 1\}$. The intact law uses the saved test counts; shifted laws retain their exact mechanism-state counts with nested context flips. At $\epsilon = 0.50$, context is independent of the mechanisms; at 1, its relation to Z_0 reverses. Reference scores remain unchanged because the reference ignores context.

D. Harm Definition

Define realized harm for case i by

$$H_i = \overline{\text{F1}}_i(0.01) - \overline{\text{F1}}_i(0.50). \tag{12}$$

The bars denote means over each case's 200 paired test batches. Harm-positive cases satisfy $H_i \geq 0.05$; labels were recomputed per learner/size stratum. F1 is $2TP/(2TP + FP + FN)$; binary balanced accuracy averages sensitivity and specificity. Localization was summarized within harm groups, and mean F1 loss across candidates. Harm-conditioned rates are descriptive because evaluation batches also defined labels.

An independent-label sensitivity check recomputed H_i from 200 fresh target batches per frozen candidate, retaining original screening batches and thresholds.

E. Score Ablations

To separate background, normalization, and cohort effects, a factorial ablation reuses all 600 $n = 10{,}000$ evaluated pairs, disjoint calibration fits, saved batches, and the same 400 paired resamples. The two backgrounds are the audit batch and each fitted model's frozen empirical 512-row training distribution. The latter is regenerated from the recorded seed and checked against its training-data hash. Normalization uses either $s_j(A)$ or raw $a_j(A)$. The comparison cohort is either other failures or healthy rows, with the own-mechanism cohort unchanged. Each variant receives its own global population calibration; the original SHAP/flip comparison also retains fixed bins. A targeted SHAP score averages $|\phi_C|$ over the entire batch and uses the same population-rank and matched-size construction. Targeted scores have access to the identity of C.

Every score is evaluated on harm-positive and below-threshold candidates under both laws. Below-threshold denotes $H_i < 0.05$, not universally harmless input use. The reference maximum score sets eligible-model warning size; the candidate localization score is evaluated at that same boundary. The additional SHAP variants use strict ranks; response scores retain independent randomized ties.

F. Process Audit

DTU Tennessee Eastman simulations supplied known fault onset and run-level units for a process-domain audit [34], [35]. Each of IDV4/IDV5 supplied 200 complete 100-hour trajectories, with 2001 timestamps and 53 inputs per run. Pairing scenarios by run number formed 200 packets. Each used the nearest samples to 16 equally spaced times over 10–25 hours and 48 over 40–95 hours per scenario; injection occurred at 30 hours. Packets were the calibration units. Partial maps were Reactor Temperature, Reactor Coolant Temp, and Reactor Coolant for IDV4; Product Sep Temp, Separator Coolant Temp, and Condenser Coolant for IDV5. Faults were mutually exclusive.

Packets 1–40 supplied training data, 41–140 supplied 100 calibration fit/packet pairs, and 141–200 supplied sixty evaluation pairs. The unaltered fitting recipe defined the null under the simulation-replicate law. Distinct pre-fault prefixes ruled out observed trajectory reuse without establishing exchangeability.

Each fit resampled 40 training packets with replacement and used a separate forest seed, preserving within-trajectory dependence. Validation trained on packets 1–30 and evaluated 31–40, comparing 64-tree bootstrap forests with square-root feature sampling and (depth, leaf minimum, class weight) settings (8, 2, balanced), (unrestricted, 2, balanced), or (unrestricted, 1, none). Balanced accuracy selected the audit model; F1 selected a predictive baseline. Interventional TreeSHAP [36] used 32 equally spaced training-background rows. To preserve scikit-learn routing, thresholds in a copied explainer forest were floored to the float32 lattice, with float32 backgrounds. Additivity against original-model probabilities was required within 10^{-6}.

G. Bias Experiments

Dose and fault-specificity comparisons tested response to a controlled injected bias. Compressor Work received 0.1, 0.25, 0.5, 1, or 2 unmodified training interquartile ranges (IQR) on all fault rows or only IDV4 rows. Label-assigned offsets were applied in training and matched testing; removal changed only the test view of the same fit. Paired refits shared training packets, forest seed, and background indices; background values reflected each fit's training offset. The response comparator replaced each feature by its training median and used unnormalized own/other-failure excess. Every dose/view retained the unaltered rank-96 cut from 100 calibration pairs at $\alpha = 0.05$. These pressure tests do not represent a naturally occurring acquisition confound.

H. Measured Load Transfer

Recorded load changes tested prediction transfer without editing sensor values. The KAIST release [37] supplied 21 normal/inner-race/outer-race recordings at 0, 2, and 4 Nm; source SHA-256 matched the publisher manifest. Thirty nonoverlapping one-second windows were spaced uniformly across each recording. Four 25.6-kHz channels were converted from MKS storage to g. Mean-centered windows supplied log RMS, log standard deviation, kurtosis, skewness, crest factor, and five Hann-periodogram power shares per channel, with band edges 0/0.5/1/2/4/12.8 kHz: forty fixed features.

Each fold held out an entire load; standardization, fitting, and selection used only the other two. Classes were equally weighted, with equal recording weights within class. Logistic regression used L-BFGS, 2000 iterations, and $C_{\mathrm{reg}} \in \{0.1, 1, 10\}$; 200-tree forests used depths $\{4, 8, \infty\}$, leaf minimum two, and square-root feature sampling. Mean macro F1 across both source-to-source directions selected parameters; ties followed listed order. Models were refitted on both source loads using seed 20261026; all six directed controls were retained. Endpoints were three-class macro F1, binary balanced accuracy, and healthy false-positive rate.

Exploratory diagnostics reconstructed the 0-Nm-only and 0+2-Nm forests. Interventional TreeSHAP [36] used thirty source-only background rows, ten equally spaced per class, to explain $P_F = P(\mathrm{inner}) + P(\mathrm{outer})$. Probability reconstruction was checked within 10^{-6}. Twenty consecutive seeds beginning at 20261026 retained fixed parameters. Collapsed three-class argmax and $\mathbb{1}\{P_F \geq 0.5\}$ were evaluated separately. Seeds and windows were not treated as independent assets.

I. Statistical Analysis

Because batches from one model are dependent, nominal-rate intervals resample 200 whole fitted-case clusters 1000 times, using shared indices across contrasts. Population intervals condition on the bank. The reported transfer and harm comparisons are exploratory, with pointwise intervals

and no multiplicity adjustment. Ancillary Holm-adjusted bin comparisons are retained in the technical materials and do not establish harm discrimination.

Retrospective localization curves match 1, 2.5, 5, 10, and 20% empirical eligible-model warnings. The matched analysis uses 400 paired resamples of calibration pairs and fitted-case clusters, recomputing ranks and boundaries while retaining each case's batches and harm label. Percentile intervals use bootstrap quantiles 0.025/0.975. The additional global warning/permutation comparisons reuse these same 400 paired draws at every rate, preserving each case's harm label and generated permutation. The extensions remain retrospective; none is a prespecified primary test. Process warning/localization intervals are exact binomial intervals conditional on the bank; separate tolerance bounds describe calibration-bank randomness.

VI. Results

A. Reference Transfer

Threshold transfer was assessed by applying fixed-model cuts to decision-equivalent refits (Fig. 1). SHAP self-warning rates were near 5%, whereas same-family refit rates were 6.0–9.6% and other-family rates were 3.7–14.1%. The logistic same-family rate at $n = 10{,}000$ was 9.62% (pointwise 95% interval 8.02–11.38%). Fixed-bin flip-excess same-family rates were 10.4–18.7%. Warning frequencies differed across probability-valued fits despite identical decisions; proxy-control warning rates were 99.75–100% under fixed-model, fixed-bin SHAP calibration.

Calibration across independent same-family fits was next compared with fixed-model calibration. Population-calibrated SHAP warning rates were 3.4–7.1% with fixed bins and 4.7–6.9% globally across six learner/size strata.

The hard-output control checked equality over the full input domain. All 3,600 reference/control comparisons agreed on every state, satisfying the conditions for equal hard-output score laws under a shared background and batch law.

B. Shortcut Localization

Matched comparisons tested whether pre-shift localization differences persisted across batch sizes, warning rates, and partitions (Fig. 2). Harm-positive counts for tree/forest/logistic models were 50/95/163 at $n = 1000$ and 46/100/150 at $n = 10{,}000$. Across sixty settings, SHAP localization was higher in 47 and flip localization in thirteen. Pointwise 95% intervals excluded zero in favor of SHAP in 24 settings and flip in four; 32 included zero. SHAP point estimates were higher in all twenty tree settings, nineteen forest settings, and eight logistic settings. Logistic models accounted for twelve of thirteen settings favoring flip. Interval counts were descriptive, without multiplicity adjustment.

Post-shift evaluation paired localization with predictive loss after context became independent. Mean F1 losses across all tree/forest/logistic candidates were 0.121, 0.260, and 0.390. False-positive rates were 11.5%, 25.0%, and 37.5%; false-negative rates remained below 0.4%. SHAP context localization was zero among harm-positive candidates under fixed-model and population calibration. At matched 5% warnings, fixed-bin flip localization was 8.4%, 43.4%, and 24.3%; the global comparison had the same direction. Shifted flip localization exceeded SHAP localization in all sixty settings, with every pointwise interval excluding zero. Predictive losses and context-localization rates therefore followed different patterns.

Warning-rate matching assessed how unequal control warning rates affected the comparison. At nominal 5% population cuts, eligible-model warnings were 4.4–6.6% for SHAP and 5.8–8.2% for flip. Matching reduced shifted forest flip localization from 71.3% to 43.4%. Randomized boundaries contributed 0.0125–2.56 percentage points of null warnings at global 5% matching.

C. Harm Screening

Attribution warnings were assessed to determine whether zero context localization also meant no warning (Table III). Before shift at $n = 10{,}000$, global SHAP warning rates on harm-positive/below-threshold candidates were 100.0/4.9% for trees, 58.7/0.3% for forests, and 7.9/0.2% for logistic models. After shift, tree rates were 35.2/4.9%, although context localization was zero; harm-positive forest and logistic warning rates were approximately 0.3% and 0.2%. Context localization and warning occurrence differed.

The known-context permutation screen assessed predictive consequences using the specified input and labels. It detected every harm-positive candidate before shift in all six learner/size strata. At $n = 10{,}000$, below-threshold warning rates were 4.98%, 5.05%, and 5.06%. Harm-positive detection equaled SHAP warnings for trees and exceeded them by 41.3 percentage points for forests (95% paired interval 33.1–49.7) and 92.1 for logistic models (87.5–95.6).

Applying the same screen after context independence yielded harm-positive warning rates of 50.3%, 50.0%, and 49.2% at $n = 10{,}000$. Detection was lower after shift than before it. Full rates and intervals at all five warning rates and both sizes were retained with the numerical tables.

Sensitivity checks assessed dependence on harm-label batches and permutation draws. Independent loss batches changed none of the 1,200 labels: harm-positive losses were at least 0.508, and below-threshold losses were at most 0.0068. Across ten additional permutation streams, pre-shift detection remained 100% in every stratum; shifted $n = 10{,}000$ detection ranged from 49.0% to 50.9%. These ranges summarized repeated randomizations, not confidence intervals.

D. Score Ablations

Group comparisons checked how often localization also occurred below the harm threshold. After shift, global flip localization was 47.0% for harm-positive forests and 25.7% below threshold; fixed-bin rates were 43.4% and 40.4%. Fixed-bin tree localization was lower above the threshold than below it (8.4% versus 27.8%). Before shift, global SHAP forest

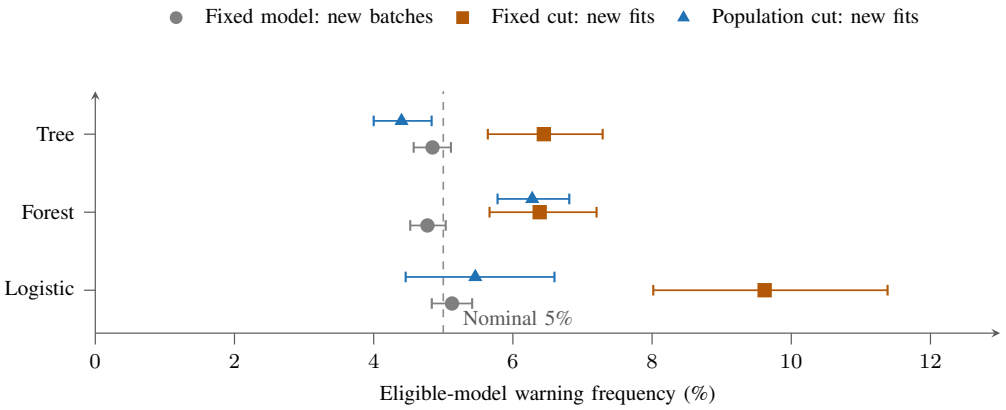


Fig. 1. **Probability-audit warning rates increased on decision-equivalent refits.** SHAP excess used fixed overlap bins and $n = 10{,}000$; refits used the same learner family. Each learner had 200 evaluated fits with 200 batches each. Whiskers are 95% fitted-case bootstrap intervals; population-cut intervals condition on the realized calibration bank. The dashed line marks nominal 5%.

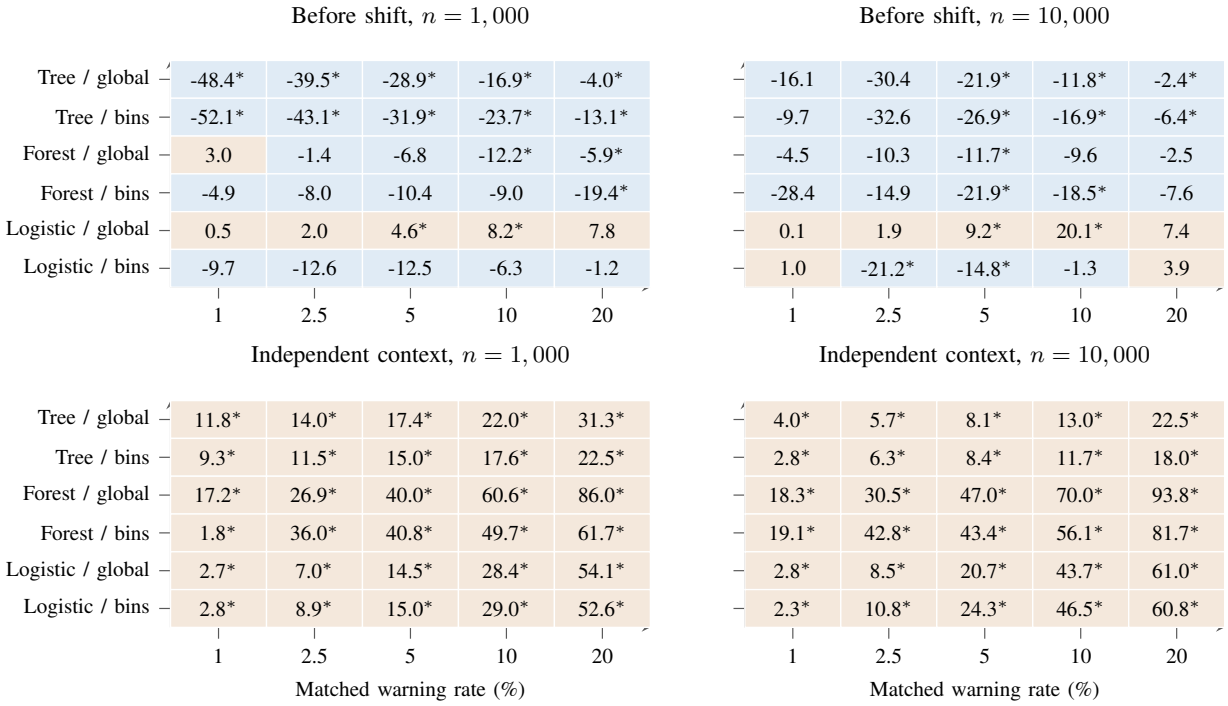


Fig. 2. **Localization differences depend on learner, warning rate, and context law.** All 120 cells show flip-minus-SHAP localization differences in percentage points across two sizes, two partitions, five matched warning rates, and two laws. Blue/negative favors SHAP; orange/positive favors flip. An asterisk marks a pointwise 95% paired-bootstrap interval excluding zero; these intervals are not multiplicity-adjusted. Fixed bins use target-law knowledge. Matching was retrospective.

localization was 58.7% versus 0.1%. Group separation differed across scores and context laws.

Factorial ablations compared background, normalization, and comparison-cohort effects. Freezing the training background alone yielded post-shift harm-positive localization of 1.6%, 1.0%, and 0% for tree/forest/logistic models. Frozen-background unnormalized SHAP yielded 6.6%, 28.6%, and 10.6%, with below-threshold rates of 12.4%, 13.5%, and 7.2%. Forest intervals were 16.9–40.8% above the harm threshold and 5.5–24.6% below it. With batch-background normalized SHAP, replacing other failures by healthy comparison rows left localization at zero in every family. Nonzero localization appeared under some background/normalization changes, while changing the comparison cohort alone left it at zero.

Targeted scores tested localization when the context identity was supplied. SHAP and flip both localized every harm-positive model, along with 52.5%, 100%, and 100% of below-threshold tree/forest/logistic candidates under both laws. Localization occurred in both groups. Complete pointwise intervals and partition comparisons were retained as reproducibility

TABLE III

HARM SCREENING AT GLOBALLY MATCHED 5% EMPIRICAL ELIGIBLE-MODEL WARNINGS, $n = 10{,}000$. ENTRIES ARE WARNING PERCENTAGES WITH POINTWISE 95% PAIRED BANK/CASE BOOTSTRAP INTERVALS. H^+ DENOTES $H_i \geq 0.05$ AND H^- DENOTES $H_i < 0.05$. SHAP AND FLIP SEARCH ALL OUTSIDE FEATURES; PERMUTATION F1 KNOWS C. COUNTS ARE 46/154, 100/100, AND 150/50. MATCHING IS RETROSPECTIVE; 100% WITH A DEGENERATE BOOTSTRAP INTERVAL IS NOT A POPULATION SENSITIVITY GUARANTEE.

Learner	Screen	Intact H^+	Intact H^-	Shifted H^+	Shifted H^-
Tree	SHAP warning	100.0 [100.0, 100.0]	4.9 [4.7, 5.2]	35.2 [31.5, 38.0]	4.9 [4.7, 5.2]
Tree	Flip warning	83.7 [78.3, 88.2]	18.5 [14.2, 24.4]	100.0 [100.0, 100.0]	18.3 [13.7, 24.2]
Tree	Permutation F1	100.0 [100.0, 100.0]	5.0 [4.9, 5.1]	50.3 [49.4, 51.5]	5.0 [4.9, 5.1]
Forest	SHAP warning	58.7 [50.3, 66.9]	0.3 [0.1, 0.7]	0.3 [0.2, 0.5]	0.5 [0.3, 0.7]
Forest	Flip warning	50.6 [39.0, 59.0]	30.4 [21.6, 40.2]	50.8 [39.2, 59.2]	30.3 [21.6, 40.2]
Forest	Permutation F1	100.0 [100.0, 100.0]	5.1 [4.8, 5.3]	50.0 [49.4, 50.5]	5.1 [4.8, 5.3]
Logistic	SHAP warning	7.9 [4.4, 12.5]	0.2 [0.0, 0.5]	0.2 [0.1, 0.3]	0.3 [0.1, 0.7]
Logistic	Flip warning	21.4 [12.7, 33.9]	16.3 [7.3, 31.0]	25.8 [16.6, 37.6]	17.3 [8.4, 31.7]
Logistic	Permutation F1	100.0 [100.0, 100.0]	5.1 [4.7, 5.5]	49.2 [48.7, 49.9]	5.1 [4.7, 5.5]

assets.

E. Process Audit

The process audit compared warnings and predictive quality under training-packet resampling. Frozen cuts were 0.03841 for SHAP and 0.02041 for response; both produced 3/60 unaltered warnings. The audit model had F1 0.803 and balanced accuracy 0.766. The F1-selected forest had F1 0.864, balanced accuracy 0.588, and a healthy-row false-positive rate of 78.5%; the all-positive baseline gave F1 0.857 and balanced accuracy 0.5. Model rankings differed between F1 and balanced accuracy.

Dose comparisons assessed localization across offset magnitude and affected fault cohort. Common-fault SHAP localization increased from 1/60 at 0.1 IQR to 11/60 at 0.25 IQR and 60/60 at 2 IQR, always through IDV5. With IDV4-only offsets, localization was zero at the three doses up to 0.5 IQR, 13/60 at 1 IQR, and 57/60 at 2 IQR, through IDV4 at the two largest doses. For common-fault offsets at 1 and 2 IQR, response localization fell from 59/60 to 44/60: eleven packets fell below the frozen cut and four lost context as a maximizing feature, with no gains. Common-fault response localization was non-monotone in dose.

Prediction replay investigated the unchanged binary F1 under IDV4-only offset removal. All 600 matched/removal metric records were reproduced from the 300 specified fits. Every fit classified all 48 IDV4 fault rows as faulty in both views at every dose, with minimum probability 0.629. Unchanged binary predictions accounted for the zero F1 differences. Full dose plots were retained with the numerical data.

F. Measured Load Transfer

Two-source fits assessed prediction transfer after source-only selection. Every fold selected $C_{\mathrm{reg}} = 0.1$ and forest depth four. Logistic macro F1 was 1.000 at every held-out load; forest macro F1 was 1.000/1.000/0.985 at 0/2/4 Nm. All six fits had zero healthy false positives under collapsed argmax.

Single-source controls examined the training/test load pairs (Fig. 3). The 0-Nm forest misclassified 26/30 healthy 4-Nm windows, versus none at 2 Nm. Every other directed forest control and all logistic controls had zero healthy errors. Healthy-row errors were confined to that forest load pair; each load supplied one normal recording and thirty evaluated windows.

Seed repeats compared the stability of both binary decision rules. At 4 Nm, the 0-Nm forest's healthy false-positive rate ranged 63.3–100% under collapsed argmax and remained 100% under $P_F \geq 0.5$. The 0+2-Nm forest had zero collapsed-argmax healthy errors across twenty seeds, but probability-threshold rates ranged 0–100%. The zero-error result did not hold across both rules. No mechanism-level warning was calibrated in this pilot.

VII. DISCUSSION

A. Calibration Scope

A warning is calibrated for a specified scalar function or distribution of functions. Hard-decision equality makes explanation transfer automatic under the same background and batch law; probability equality is a stronger condition. Our transfer inflation quantifies the consequence of overlooking that distinction. Population calibration addresses fitting variation only for the declared training procedure and eligibility gate. Cross-family rates can lie below or above nominal because their score laws differ; there is no expected direction of transfer error.

Case bootstraps condition on the bank, while the paired analysis resamples both bank and evaluated fits. Order-statistic tolerance bounds describe a different uncertainty, the realized cut's tail probability. Matching observed warning frequencies is useful for retrospective comparisons but does not certify a fixed deployed threshold. Target-informed overlap bins illustrate sensitivity to generator knowledge; source-derived partitions and global calibration are the deployable alternatives evaluated here.

B. Warning Interpretation

Outside-set excess asks whether attribution is concentrated on one mechanism relative to other failures. It neither directly counts healthy-row errors nor guarantees localization of shared context reliance. Healthy-row exclusion alone is not a universal insensitivity theorem: interventional SHAP can query healthy hybrid states, and its background can include healthy

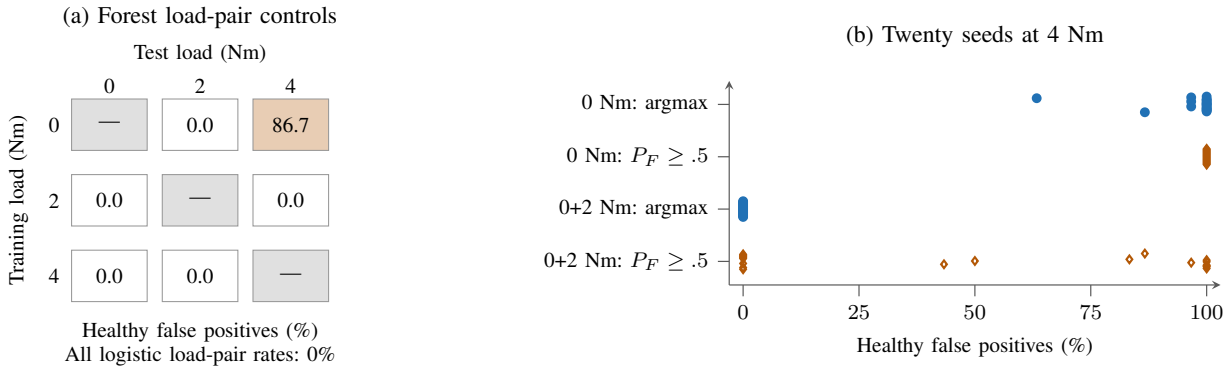


Fig. 3. **Measured healthy-row failure rates varied with training set and decision rule.** (a) Healthy false-positive rates for all six directed forest load pairs at the original seed; every logistic control was zero. Diagonal cells were not evaluated. (b) Each point denotes one of twenty forest seeds on the same 4-Nm normal recording; vertical jitter separates seeds. Labels specify source loads and the binary decision rule. Argmax collapses the three-class prediction; P_F sums fault probabilities. Points are not independent-asset replicates or confidence intervals.

observations. A finite-domain counterexample changing only one healthy-state prediction changes the failure-row attributions and excesses; full values accompany the technical materials. Cohort coverage must therefore be assessed together with the output, background, normalization, and signed contrast.

The logistic crossover is sensitive to audit design. For global calibration at $n = 10{,}000$ and 5% matched size, removing normalization raised SHAP localization by 10.1 percentage points (paired 95% interval 7.1–13.0); freezing the training background raised it by 28.2 (20.5–36.0), reversing the flip comparison. Logit reconstruction on all 400 logistic candidates confirmed affine logits: their feature-flip responses are constant, so raw own/other-failure flip excess is zero. Probability outputs introduce state dependence. These paired effects demonstrate dependence on output, normalization, background, and calibration; they do not identify one cause for every grid crossover.

When a suspected input and labels are available, permutation-performance screening is the more direct check of that intervention's predictive consequence. Its post-shift score addresses an additional marginal permutation; it does not estimate loss relative to the intact environment. When the input's identity is unknown, an attribution warning provides a candidate for investigation. Neither screen establishes a physically valid intervention or a maintenance action without further validation.

C. Limitations

The Boolean world supplies its mechanisms, relevance map, and eligible training population. Primary harm labels use the screening batches; a fresh-batch sensitivity check preserved all labels but does not constitute independent field validation. The observed gap between loss groups leaves sensitivity near the 0.05 harm threshold untested. The baseline knows context and uses one permutation per batch in the primary comparison. Its bootstrap retains those permutations; ten separate streams assess randomization sensitivity. The many rate and design comparisons are exploratory, with pointwise rather than simultaneous intervals. They should not be interpreted as an independently confirmed harm detector.

DTU uses simulated disturbances and deliberately label-assigned offsets. IDV4's saturated decisions explain its binary-loss insensitivity. KAIST supplies measured load-transfer failures, but native function-creation dates are perfectly class-coupled and are not authenticated acquisition dates. Comparing one- and two-source fits changed both load coverage and training sample size. Independent bearing identities and an exhaustive physical feature map remain unverified. The load pilot therefore establishes neither causal acquisition shortcuts nor calibrated warning sensitivity. Neither study establishes mild-bias detection or new-plant transfer. The study evaluates tabular explanations, not temporal attribution or large-scale monitoring.

D. Reproducibility

The reproducibility package includes dependencies, seeds, input hashes, probability tables, audit counts, generated permutations, bootstrap draws, and prediction-replay records. Protocol history distinguishes original settings from retrospective extensions. The driver regenerates the numerical comparisons, native TikZ figures, and manuscript checks. Complete overlap, map-bound, and secondary measured-data details are retained outside the paper in the technical materials.

VIII. Conclusion

Mechanism-level attribution warnings require an explicit output and reference population. In this controlled study, probability explanations vary across models whose hard decisions agree, while hard-output explanations agree under the same background and batch law. Localization differences depend on learner and warning rate. When context is known, performance screening assesses predictive consequences more directly than flip-excess localization. Measured load transfer additionally exposed healthy-row errors and decision-rule sensitivity. Engineers should calibrate over the fitting variation they intend to cover, distinguish any outside-feature warning from localization of a specified input, and validate predictive

harm with labels and a justified intervention. The process and measured-load illustrations support these interpretation boundaries, rather than a claim of field-ready fault detection.

Acknowledgment

Claude (Anthropic) and Codex (OpenAI) assisted prose revision and analysis-code drafting, including experimental and verification scripts. The authors are responsible for the scientific claims and verification. The project repository is https://github.com/kowshik24/glowing-broccoli.

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